PC Web Page Programming

1. Web Page Programming Instructions

PC web page programming uses the Jupyter lab platform, calling built-in buttons and sliders and other controls to generate sliders for controlling the six joints and movement speed of the robotic arm. Each time you drag the slider, you can control the corresponding servo of the robotic arm to rotate.

We provide a reference program PC_Control.ipynb in the source code folder. You need to copy this program to the container's jupyter-lab and place the program in the **~ directory of the motherboard**. Enter the following command in the terminal on the motherboard to copy the program to the docker container:

```
docker cp ~/PC_Control.ipynb container_name_or_container_ID:/root
```

The container name and container ID need to be replaced with your own started container name or container ID, which can be viewed through docker ps.

2. Important Code Explanation

Enter the docker container and enter the following command to start jupyter-lab:

```
cd
jupyter-lab --allow-root
```

Then enter ip:9999 in the web browser to enter the jupyter-lab interface in docker.

```
#Web control program source code path, view in docker,
/root/PC_Control.ipynb
```

Create three buttons for controlling robotic arm reset, power on, and power off functions.

```
def on_button_clicked(b):
    with output:
        print("Button clicked:", b.description)
    if b.description == 'Reset':
        reset_joints()
    elif b.description == 'Power_on':
        Arm.Arm_serial_set_torque(True)
        b.icon = 'check'
        button_power_off.icon = 'uncheck'
    elif b.description == 'Power_off':
        Arm.Arm_serial_set_torque(False)
        b.icon = 'check'
        button_power_on.icon = 'uncheck'
```

Create six sliders for controlling the six joints of the robotic arm.

```
def on_slider_S1(angle):
```

```
    print("J1:", angle)
    mc.send_angle(1, angle, g_speed)
def on_slider_S2(angle):
    print("J2:", angle)
    mc.send_angle(2, angle, g_speed)
def on_slider_S3(angle):
    print("J3:", angle)
    mc.send_angle(3, angle, g_speed)
def on_slider_S4(angle):
    print("J4:", angle)
    mc.send_angle(4, angle, g_speed)
def on_slider_S5(angle):
    print("J5:", angle)
    mc.send_angle(5, angle, g_speed)
def on_slider_S6(angle):
    print("J6:", angle)
    mc.send_angle(6, angle, g_speed)
```

Create a slider for controlling the movement speed of the servo.

```
slider_speed = widgets.IntSlider(description='Speed:', value=1000, min=0,
max=2000, step=100, orientation='horizontal')

def on_slider_speed(value):
    global g_speed
    g_speed = value
    print("speed:", value)

widget_speed = widgets.interactive(on_slider_speed, value=slider_speed)
```

Reset the joint angles of the robotic arm.

```
def reset_joints():
    if button_power_off.icon == 'check':
        Arm.Arm_serial_set_torque(True)
        time.sleep(1)
    Arm.Arm_serial_servo_write6_array([90, 164, 18, 0, 90, 30], 1000)
    slider_S1.value = 90
    slider_S2.value = 164
    slider_S3.value = 18
    slider_S4.value = 0
    slider_S5.value = 90
    slider_S6.value = 30
    slider_speed.value = 1000
    button_power_on.icon = 'check'
    button_power_off.icon = 'uncheck'
```

Create a camera display window to read and display camera images in real time.

```
imgbox = widgets.Image(format='jpg', width=640, height=480,
layout=widgets.Layout(align_self='center'))
model = 'Start'
```

```
def camera():
    global model
```

```
    capture = cv.VideoCapture(0)
    capture.set(cv.CAP_PROP_FRAME_WIDTH, 640)
    capture.set(cv.CAP_PROP_FRAME_HEIGHT, 480)
    while capture.isOpened():
        try:
            _, img = capture.read()
            if model == 'Exit':
                break
            imgbox.value = cv.imencode('.jpg', img)[1].tobytes()
        except:
            break
    with output:
        print("capture release")
    capture.release()
```

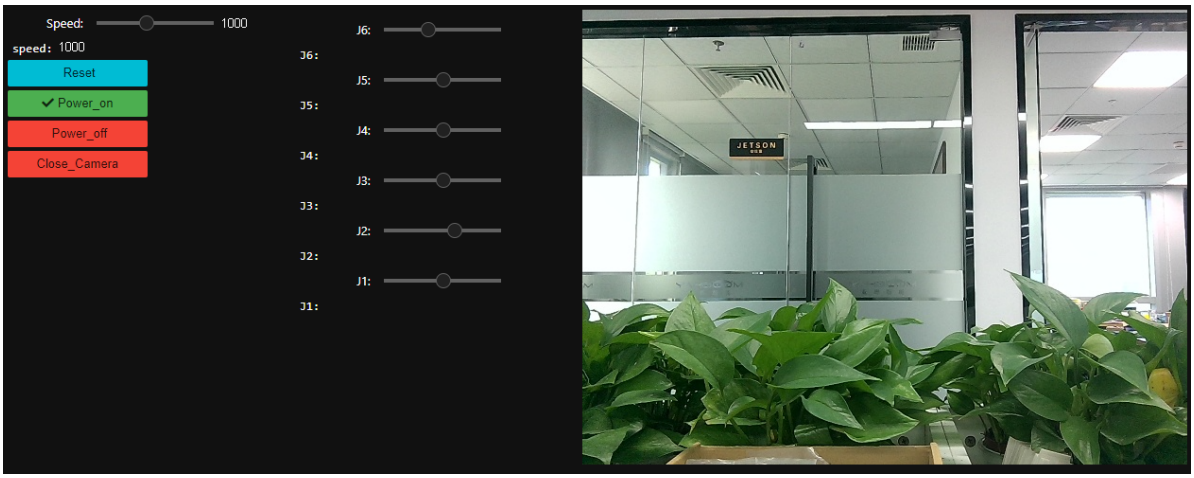
Create a close button to end the program and release resources.

```
button_close = widgets.Button(description='Close_Camera', button_style='danger')
def button_close_Callback(value):
    global model
    model = 'Exit'
    with output: print(model)
button_close.on_click(button_close_Callback)
```

3. Running the Program

Click the "Run Entire Program" button on the jupyterlab toolbar, then scroll to the bottom.

You can see the relevant control controls on the left and the camera display on the right.



The corresponding positions of the robotic arm joints are shown in the figure below.